Scenario	Design		Composition		Networks			Statistical information				Numerical conditioning	
	Cells per bulk n_i	Cell-size ratio	Shannon $H^\star$	$\|\|\boldsymbol{p}\|\|_2^2$	Edge density	Mean degree	Hoyer sparsity	Mean overlap $\bar{\omega}$	Max overlap $\omega_{\max}$	Hellinger	max f_{cov}	$\kappa\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$	$\lambda_{\min}\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$
G = 20 / 1000 cells per bulk / balanced / low sibling overlap													
a = 1 / equal sizes	1e+03	1	1	0.25	0.1	1.9	0.52	0.00346	0.0175	0.994	0.121	4.3	0.000213
a = 1 / sizes 1:1:5:5	1e+03	5	1	0.25	0.1	1.9	0.611	0.00308	0.0152	0.994	0.223	5.71	0.0025
a = 5 / equal sizes	1e+03	1	1	0.25	0.1	1.9	0.52	0.00333	0.0142	0.994	0.121	4.3	0.000213
a = 5 / sizes 1:1:5:5	1e+03	5	1	0.25	0.1	1.9	0.611	0.00312	0.0148	0.994	0.223	5.71	0.0025
G = 20 / 1000 cells per bulk / balanced / high sibling overlap													
a = 1 / equal sizes	1e+03	1	1	0.25	0.1	1.9	0.52	0.0266	0.106	0.962	0.252	4.3	0.00052
a = 1 / sizes 1:1:5:5	1e+03	5	1	0.25	0.1	1.9	0.611	0.0264	0.109	0.963	0.411	5.71	0.00608
a = 5 / equal sizes	1e+03	1	1	0.25	0.1	1.9	0.52	0.0271	0.112	0.962	0.252	4.3	0.00052
a = 5 / sizes 1:1:5:5	1e+03	5	1	0.25	0.1	1.9	0.611	0.0267	0.11	0.963	0.411	5.71	0.00608
G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap													
a = 1 / equal sizes	1e+03	1	0.861	0.34	0.1	1.9	0.438	0.00417	0.0198	0.994	0.413	5.2	0.000284
a = 1 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.1	1.9	0.43	0.00408	0.0202	0.994	0.372	7.23	0.00311
a = 5 / equal sizes	1e+03	1	0.861	0.34	0.1	1.9	0.438	0.00454	0.023	0.994	0.413	5.2	0.000284
a = 5 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.1	1.9	0.43	0.00442	0.021	0.994	0.372	7.23	0.00311
G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap													
a = 1 / equal sizes	1e+03	1	0.861	0.34	0.1	1.9	0.438	0.0349	0.141	0.962	0.631	5.2	0.000691
a = 1 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.1	1.9	0.43	0.0343	0.144	0.963	0.59	7.23	0.00756
a = 5 / equal sizes	1e+03	1	0.861	0.34	0.1	1.9	0.438	0.0353	0.142	0.962	0.631	5.2	0.000691
a = 5 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.1	1.9	0.43	0.0341	0.143	0.963	0.59	7.23	0.00756
G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap													
a = 1 / equal sizes	1e+05	1	1	0.25	0.1	1.9	0.52	0.00288	0.013	0.994	0.121	4.3	0.000213
a = 1 / sizes 1:1:5:5	1e+05	5	1	0.25	0.1	1.9	0.611	0.00346	0.0152	0.994	0.223	5.71	0.0025
a = 5 / equal sizes	1e+05	1	1	0.25	0.1	1.9	0.52	0.00383	0.0168	0.994	0.121	4.3	0.000213
a = 5 / sizes 1:1:5:5	1e+05	5	1	0.25	0.1	1.9	0.611	0.00317	0.0142	0.994	0.223	5.71	0.0025
G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap													
a = 1 / equal sizes	1e+05	1	1	0.25	0.1	1.9	0.52	0.0274	0.108	0.962	0.252	4.3	0.00052
a = 1 / sizes 1:1:5:5	1e+05	5	1	0.25	0.1	1.9	0.611	0.026	0.105	0.963	0.411	5.71	0.00608
a = 5 / equal sizes	1e+05	1	1	0.25	0.1	1.9	0.52	0.0276	0.11	0.962	0.252	4.3	0.00052
a = 5 / sizes 1:1:5:5	1e+05	5	1	0.25	0.1	1.9	0.611	0.0275	0.111	0.963	0.411	5.71	0.00608
G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap													
a = 1 / equal sizes	1e+05	1	0.861	0.34	0.1	1.9	0.438	0.00454	0.021	0.994	0.413	5.2	0.000284
a = 1 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.1	1.9	0.43	0.00438	0.0202	0.994	0.372	7.23	0.00311
a = 5 / equal sizes	1e+05	1	0.861	0.34	0.1	1.9	0.438	0.00362	0.0192	0.994	0.413	5.2	0.000284
a = 5 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.1	1.9	0.43	0.00425	0.0205	0.994	0.372	7.23	0.00311
G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap													
a = 1 / equal sizes	1e+05	1	0.861	0.34	0.1	1.9	0.438	0.0345	0.141	0.962	0.631	5.2	0.000691
a = 1 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.1	1.9	0.43	0.034	0.142	0.963	0.59	7.23	0.00756
a = 5 / equal sizes	1e+05	1	0.861	0.34	0.1	1.9	0.438	0.0335	0.139	0.962	0.631	5.2	0.000691
a = 5 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.1	1.9	0.43	0.0338	0.138	0.963	0.59	7.23	0.00756
G = 200 / 1000 cells per bulk / balanced / low sibling overlap													
a = 1 / equal sizes	1e+03	1	1	0.25	0.0499	9.94	0.419	0.00329	0.0198	0.995	0.523	7.92	2.63e-05
a = 1 / sizes 1:1:5:5	1e+03	5	1	0.25	0.0499	9.94	0.76	0.00358	0.021	0.995	0.821	10.1	0.000361
a = 5 / equal sizes	1e+03	1	1	0.25	0.0499	9.94	0.419	0.00325	0.019	0.995	0.523	7.92	2.63e-05
a = 5 / sizes 1:1:5:5	1e+03	5	1	0.25	0.0499	9.94	0.76	0.00321	0.019	0.995	0.821	10.1	0.000361
G = 200 / 1000 cells per bulk / balanced / high sibling overlap													
a = 1 / equal sizes	1e+03	1	1	0.25	0.0499	9.94	0.419	0.0272	0.152	0.965	0.847	7.92	0.000133
a = 1 / sizes 1:1:5:5	1e+03	5	1	0.25	0.0499	9.94	0.76	0.0265	0.148	0.965	0.959	10.1	0.00182
a = 5 / equal sizes	1e+03	1	1	0.25	0.0499	9.94	0.419	0.0265	0.148	0.965	0.847	7.92	0.000133
a = 5 / sizes 1:1:5:5	1e+03	5	1	0.25	0.0499	9.94	0.76	0.0245	0.134	0.965	0.959	10.1	0.00182
G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap													
a = 1 / equal sizes	1e+03	1	0.861	0.34	0.0499	9.94	0.398	0.00433	0.0242	0.995	0.89	9.04	3.5e-05
a = 1 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.0499	9.94	0.712	0.00425	0.0245	0.995	0.921	12.8	0.000473
a = 5 / equal sizes	1e+03	1	0.861	0.34	0.0499	9.94	0.398	0.00425	0.0245	0.995	0.89	9.04	3.5e-05
a = 5 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.0499	9.94	0.712	0.00379	0.0215	0.995	0.921	12.8	0.000473
G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap													
a = 1 / equal sizes	1e+03	1	0.861	0.34	0.0499	9.94	0.398	0.0355	0.198	0.965	0.976	9.04	0.000177
a = 1 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.0499	9.94	0.712	0.0342	0.192	0.965	0.983	12.8	0.00239
a = 5 / equal sizes	1e+03	1	0.861	0.34	0.0499	9.94	0.398	0.0364	0.202	0.965	0.976	9.04	0.000177
a = 5 / sizes 1:1:5:5	1e+03	5	0.861	0.34	0.0499	9.94	0.712	0.0358	0.201	0.965	0.983	12.8	0.00239
G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap													
a = 1 / equal sizes	1e+05	1	1	0.25	0.0499	9.94	0.419	0.00233	0.0135	0.995	0.523	7.92	2.63e-05
a = 1 / sizes 1:1:5:5	1e+05	5	1	0.25	0.0499	9.94	0.76	0.00279	0.0158	0.995	0.821	10.1	0.000361
a = 5 / equal sizes	1e+05	1	1	0.25	0.0499	9.94	0.419	0.00333	0.0198	0.995	0.523	7.92	2.63e-05
a = 5 / sizes 1:1:5:5	1e+05	5	1	0.25	0.0499	9.94	0.76	0.00325	0.019	0.995	0.821	10.1	0.000361
G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap													
a = 1 / equal sizes	1e+05	1	1	0.25	0.0499	9.94	0.419	0.0281	0.158	0.965	0.847	7.92	0.000133
a = 1 / sizes 1:1:5:5	1e+05	5	1	0.25	0.0499	9.94	0.76	0.0274	0.151	0.965	0.959	10.1	0.00182
a = 5 / equal sizes	1e+05	1	1	0.25	0.0499	9.94	0.419	0.0268	0.149	0.965	0.847	7.92	0.000133
a = 5 / sizes 1:1:5:5	1e+05	5	1	0.25	0.0499	9.94	0.76	0.0276	0.153	0.965	0.959	10.1	0.00182
G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap													
a = 1 / equal sizes	1e+05	1	0.861	0.34	0.0499	9.94	0.398	0.00454	0.0265	0.995	0.89	9.04	3.5e-05
a = 1 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.0499	9.94	0.712	0.00442	0.0255	0.995	0.921	12.8	0.000473
a = 5 / equal sizes	1e+05	1	0.861	0.34	0.0499	9.94	0.398	0.00388	0.023	0.995	0.89	9.04	3.5e-05
a = 5 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.0499	9.94	0.712	0.00342	0.0198	0.995	0.921	12.8	0.000473
G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap													
a = 1 / equal sizes	1e+05	1	0.861	0.34	0.0499	9.94	0.398	0.0355	0.198	0.965	0.976	9.04	0.000177
a = 1 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.0499	9.94	0.712	0.0335	0.188	0.965	0.983	12.8	0.00239
a = 5 / equal sizes	1e+05	1	0.861	0.34	0.0499	9.94	0.398	0.035	0.192	0.965	0.976	9.04	0.000177
a = 5 / sizes 1:1:5:5	1e+05	5	0.861	0.34	0.0499	9.94	0.712	0.0345	0.193	0.965	0.983	12.8	0.00239

Mean signature: $\kappa(\boldsymbol{\mu}) = 3.46$ to 15, $\lambda_{\min}(\boldsymbol{\mu}^\top\boldsymbol{\mu}) = 0.00167$ to 0.0167. (range over the scenarios)
Green: easiest value of a column; red: hardest. Easiest is the highest number of cells, Shannon entropy, Hoyer sparsity, Hellinger distance, covariance share and $\lambda_{\min}$; the lowest cell-size ratio, concentration, edge density, mean degree, overlap and condition number. Shading and size: Shannon entropy, mean overlap and condition number.